

Companion to Bird & de Moor, Algebra of Programming

§8. Thinning Algorithms

§8.1. Thinning

(8.1)

For $Q : A \rightarrow A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\in \setminus (Q^\circ \in)) : EA \rightarrow EA$ (8.1).

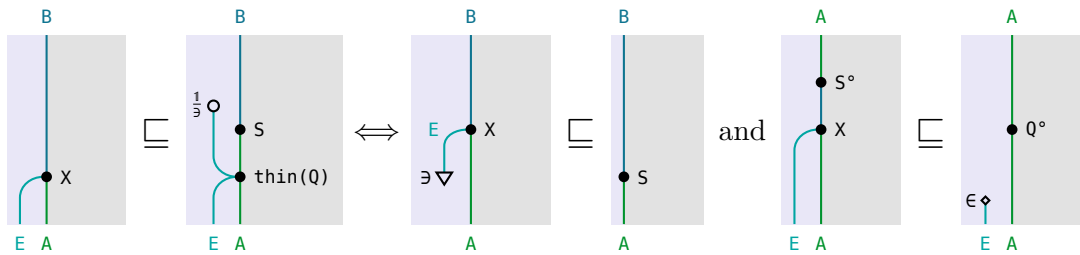
$ys \text{ thin}(Q) xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: $\in X$ runs $a \rightarrow ys \rightarrow xs$, member of ys first, and $Q^\circ \in$ runs $a \rightarrow b \rightarrow xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q^\circ \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(8.1a)

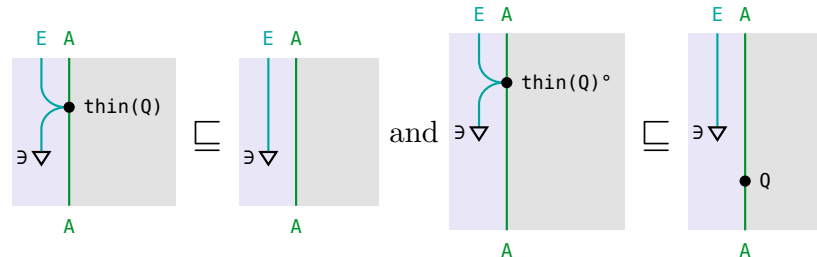
the law	what it says
$X \in \frac{S}{\exists} \text{ thin}(Q) \iff X \in S$ and $S^\circ X \in Q^\circ \in$	everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones
$\text{thin}(Q) \exists \in \exists$	everything a thinning keeps was in the set
$\text{thin}(Q)^\circ \exists \in \exists Q$	every element of the set has a Q -lower bound among the kept ones
$Q \in R \implies \text{thin}(Q) \subseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
$1 \in Q \implies 1 \in \text{thin}(Q)$ $QQ \in Q \implies \text{thin}(Q) \text{ thin}(Q) \subseteq \text{thin}(Q)$	keeping everything is always a legal thinning
$Q \in R \wedge 1 \in Q \wedge RR \in R \implies \text{est}(R) = \text{thin}(Q) \text{ est}(R)$ $Q \in R, 1 \in Q, RR \in R$ — (8.1.2a); weaker than the book's "both preorders": Q transitive is never used and $1 \in R$ follows	thin-introduction: thinning first cannot lose an R -minimum
$\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\exists}$ (8.2) — (8.2)	thin-elimination: keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
$\frac{S}{\exists} \text{ thin}(Q) \supseteq \frac{S}{\exists} \text{ est}(R) \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \subseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
$P(\text{thin}(Q)) \text{ union} \in \text{union thin}(Q)$ (8.4)	thinning each member set is a thinning of the union

§8.1.1. $X \in (S^\circ \exists) \text{ thin}(Q) \iff X \in S$ and $S^\circ X \in Q^\circ \in$

(8.1.1a)



(8.1.1b)



(8.1.1a) at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \subseteq \text{thin}(Q)$, since $\exists \exists = \Lambda(\exists) = 1$

§8.1.2. $\text{est}(R) = \text{thin}(Q) \text{ est}(R)$ given $Q \in R, 1 \in Q, RR \in R$

$\text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R)$ (8.1.2a)

the $\sqsubseteq$ half: keeping everything is a thinning — $1 \sqsubseteq Q$

(a) $1 \sqsubseteq \text{thin}(Q)$ — (8.1.1a)

$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists$ (8.1.2b)

the $\sqsubseteq$ half, first condition of the UP of est at $X \triangleq \text{thin}(Q) \text{ est}(R)$ — est-up

(a) $\text{est}(R) \sqsubseteq \exists$ — est-laws

(b) $\text{thin}(Q) \sqsubseteq \exists$ — (8.1.1a)

$\in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$ (8.1.2c)

the $\sqsubseteq$ half, second condition — $Q \sqsubseteq R$, R transitive

(a) $\in \text{thin}(Q) \sqsubseteq Q^\circ \in$ — (8.1.1a)

(b) $\in \text{est}(R) \sqsubseteq R^\circ$ — est-laws

(c) R transitive

$\text{est}(Q) \frac{1}{\exists} \sqsubseteq \text{thin}(Q)$ (8.2)

the singleton holding a Q -least member of a set is a thinning of that set

$\text{est}(Q) \frac{1}{\exists} \sqsubseteq \exists$

every member of the singleton is a member of the set

(a) $\triangleq \text{thin}, \exists/\exists$ half

(b) $\frac{1}{\exists} \sqsubseteq 1$ — pow-laws; $\text{est}(Q) \sqsubseteq \exists$ — est-laws

$\in \text{est}(Q) \frac{1}{\exists} \sqsubseteq Q^\circ \in$

the singleton's member stands in Q to every member of the set

(a) $\triangleq \text{thin}, \in \setminus (Q^\circ \in)$ half

(b) $\in \text{est}(Q) \sqsubseteq Q^\circ$ — est-up

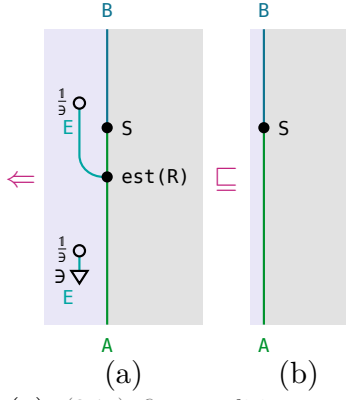
(c) $\frac{1}{\exists} \sqsubseteq \in$ — pow-laws

$$\text{RnS}^\circ \text{S} \sqsubseteq \text{Q} \Rightarrow \frac{\text{S}}{\exists} \text{est}(\text{R}) \frac{1}{\exists} \sqsubseteq \frac{\text{S}}{\exists} \text{thin}(\text{Q})$$

(8.3)

given $\text{Rn}(\text{S}^\circ \text{S}) \sqsubseteq \text{Q}$, Q a preorder

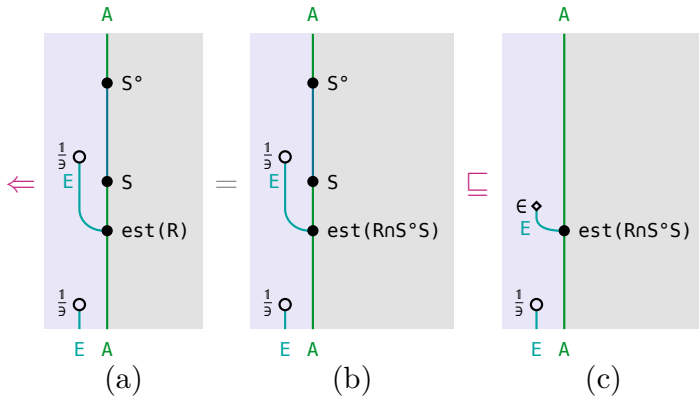
$$\frac{\text{S}}{\exists} \text{est}(\text{R}) \frac{1}{\exists} \exists \sqsubseteq \text{S}$$

every member of the singleton is a value S returns

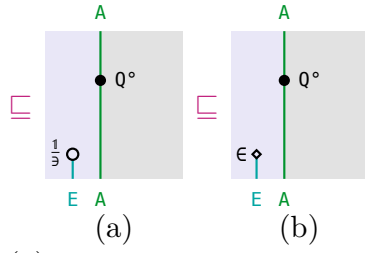
(a) (8.1a), first condition

(b) $\frac{1}{\exists} \exists = 1$ — pow-laws; $\frac{\text{S}}{\exists} \text{est}(\text{R}) \sqsubseteq \text{S}$ — est-laws

$$\text{RnS}^\circ \text{S} \sqsubseteq \text{Q} \Rightarrow \text{S}^\circ \frac{\text{S}}{\exists} \text{est}(\text{R}) \frac{1}{\exists} \sqsubseteq \text{Q}^\circ \in$$

the singleton's member stands in Q to every value S returns at the same argument

(a) (8.1a), second condition

(b) $\frac{\text{S}}{\exists} \text{est}(\text{R}) = \frac{\text{S}}{\exists} \text{est}(\text{RnS}^\circ \text{S})$ — est-laws(c) $\text{S}^\circ \frac{\text{S}}{\exists} \sqsubseteq \in$ — pow-laws(a) $\in \text{est}(\text{RnS}^\circ \text{S}) \sqsubseteq (\text{RnS}^\circ \text{S})^\circ$ — est-up; $\text{Rn}(\text{S}^\circ \text{S}) \sqsubseteq \text{Q}$ (b) $\frac{1}{\exists} \sqsubseteq \in$ — pow-laws

Theorem 8.1

$$Q \sqsubseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \sqsubseteq \langle \frac{(S)}{\exists} \text{thin}(Q) \rangle$$

thinning at every step of the reduce is a thinning of the whole candidate set — S monotonic on Q , Q a preorder

(8.1a) at $X \triangleq \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle$, $\langle (S) \rangle$ for its S

(a) fusion, hylomorphism theorem
 (b) $S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$ — mon-str, relator-laws
 (c) $S^\circ F(\epsilon) \sqsubseteq \epsilon$ — pow-laws
 (d) $\epsilon \text{thin}(Q) \sqsubseteq Q^\circ \epsilon$ — $\triangleq \text{thin}$, adj-all

Corollary 8.1

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R \sqsubseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \langle \frac{(S)}{\exists} \text{est}(R) \rangle$$

the thinning fold refines the optimisation problem itself — S monotonic on Q , $Q \sqsubseteq R$, both preorders

(a) thinning theorem
 (b) $\text{est}(R) = \text{thin}(Q) \text{est}(R)$ — (8.1a)

§8.2. Paths in a layered network

(8.2a)

$F(A, X) = A + A \times X$, $L = \text{list}^+$ with initial algebra $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$.
 $\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$, $\text{consw}(a, (xs, n)) = (\text{cons}(a, xs), \text{wt}(a, \text{head}(xs)) + n)$.
 $\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$, $\langle [\text{wrapz}, \text{consw}] \rangle = \langle 1, \text{cost} \rangle$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.
 $Q \triangleq R \circ (\text{head} \text{head}^\circ)$, $S \triangleq F(1, \exists) \alpha$, $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$, $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$, $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$.

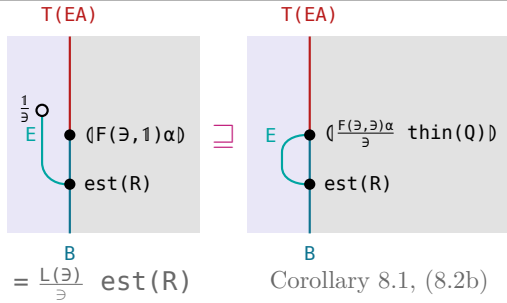
(8.2b)

the law	what it says
$F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
$S \text{head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{head}$ is simple, which gives $R \circ (S^\circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge QQ \sqsubseteq Q \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(F(\exists, 1)\alpha, Q) \wedge Rn(F(1, \exists)\alpha)^\circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \downarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \quad (8.2c)$$

$$\text{est}(R)) \downarrow \text{est}(R) \sqsubseteq \frac{F(\exists, 1)\alpha}{\exists} \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers



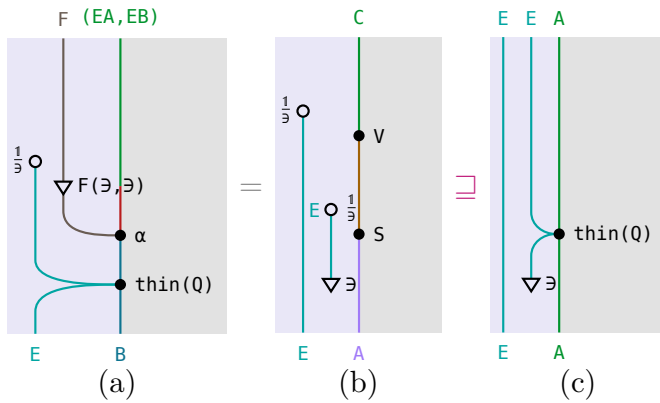
$\sqsubseteq$ (8.2d) under $\downarrow$ monotonic: the whole chain runs inside the reduce, and the $\text{est}(R)$ behind it never moves.

$$Rn(F(1, \exists)\alpha)^\circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{est}(R) \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q) \quad (8.2c)$$

thinning the algebra of a layered network costs no more than taking the program's two cases

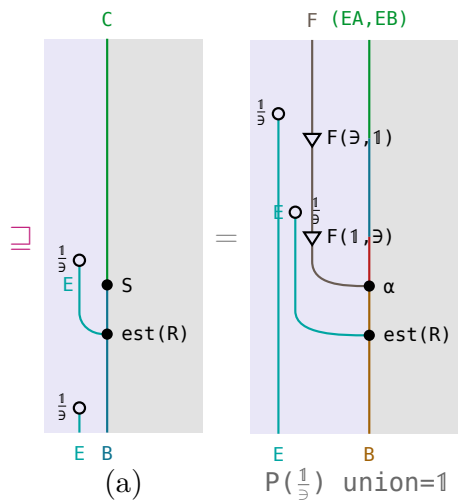
$$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union} \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q)$$

thinning each set the algebra's parts build, then taking the union, is below thinning the one set the whole algebra builds



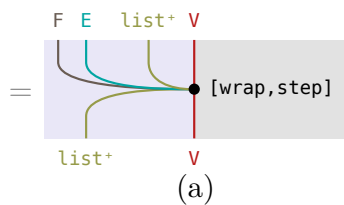
$$Rn(F(1, \exists)\alpha)^\circ F(1, \exists)\alpha \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{est}(R) \sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) P(\text{thin}(Q)) \text{union}$$

choosing an R-least member of each of those sets is below thinning each and taking the union



$$\text{cp } P(\alpha) \text{est}(\text{path}) = [\text{wrap}, \text{step}]$$

at the layered network, choosing an R-least path from each case is the program's two cases $[\text{wrap}, \text{step}]$



$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \text{ — } \triangle \text{ cost}$$

§8.3. Implementing thin

(8.3a)

$\text{setify} : [A] \rightarrow EA$, $\text{cup} : EA \times EA \rightarrow EA$, $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$, $\text{listcp} : F(L) \rightarrow LF$, $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P$ for P a connected preorder.

$\text{thinlist}(Q)$ is any $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ with $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$; one is $\llbracket \text{nil}, \text{bump } Q \rrbracket$, $\text{bump } Q (a, []) = [a]$, $\text{bump } Q (a, [b] \# xs) = (b \sqsubseteq a \rightarrow [a] \# xs, a \sqsubseteq b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; P a connected preorder with both f_1, f_2 monotonic on P ; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$.

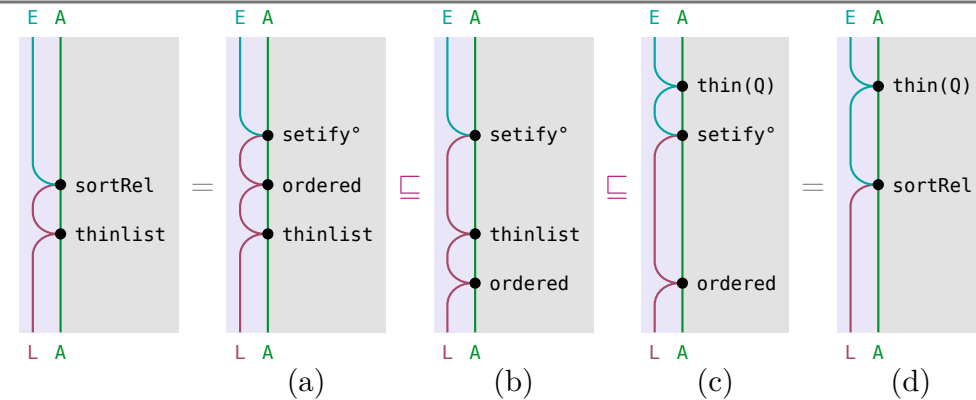
(8.3b)

the law	what it says
$\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$ (8.5), Q connected, xs non-empty	what thinning should come to when it can: one element
$\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P)$ (8.6) — (8.6)	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
$\text{sort}(P) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)	a minimum of the sorted list is a minimum of the set
$\text{sort}(f P f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(P)$ (8.8)	shunt a function through a sort
$\text{sort}(P) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(P)$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set
$(\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P)$ (8.10)	merging two sorted lists sorts their union
$F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(FP)$ (8.11), F linear	listcp is the list implementation of the cartesian product $\text{cp}(F)$

Map(setify) $\wedge$ Coreflexive(ordered) $\wedge$ thinlist $\sqsubseteq \text{subseq} \wedge$ ordered subseq $\sqsubseteq \text{subseq ordered} \wedge$ thinlist (8.6)

setify $\sqsubseteq \text{setify thin}(Q) \Rightarrow \text{sortRel thinlist} \sqsubseteq \text{thin}(Q) \text{ sortRel}$

a thinning of the sorted list lists a thinning of the set — P a connected preorder, $\text{thinlist}(Q) \sqsubseteq \text{subseq}$.



(a) $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$

(b) $\text{ordered } P \text{ thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ ordered } P \text{ — } \triangleq \text{thinlist}$

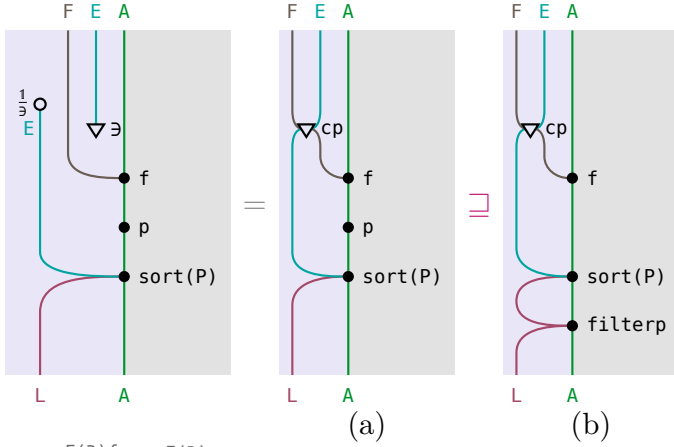
(c) $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q) \text{ — } \triangleq \text{thinlist}$, dom-laws, triple-chains

(d) $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ — } \triangleq \text{thinlist}$

Lemma 8.1

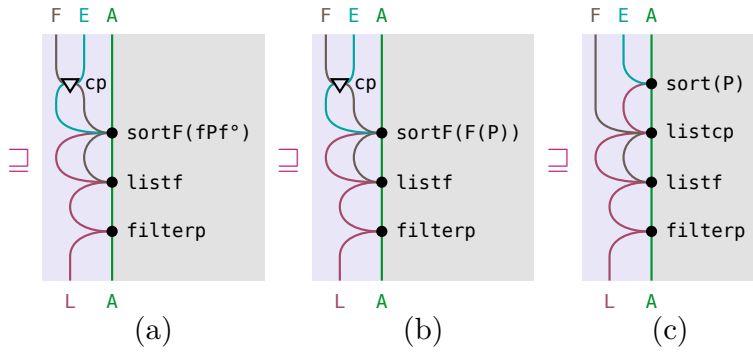
$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge$
 $\text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}$
 $\text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$
 one sorted list built from sorted arguments, instead of a set built and then sorted — $f : FA \rightarrow A$ monotonic
 on P , p coreflexive, F linear.

$\text{Map}(f) \wedge \text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P) \Rightarrow \text{cp } P(f) \text{sort}(P) \text{filterp} \sqsubseteq \frac{F(\exists)fp}{\exists} \text{sort}(P)$
 sorting the f -images of the product and filtering by p is below sorting the set $F(\exists)fp$ builds



- (a) $\frac{F(\exists)fp}{\exists} = \frac{F(\exists)}{\exists} E(fp)$ — pow-laws, $\triangleq$ thinlist
 $E(f) = P(f)$, f a map — powrel-laws
 (b) $\text{sort}(P) \text{filterp} \sqsubseteq E(p) \text{sort}(P)$ — (8.3b)

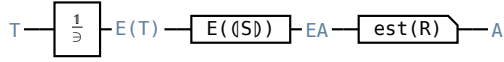
$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f, P) \wedge \text{sortF}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P) \wedge$
 $F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp} \text{sortF}(F(P)) \Rightarrow F(\text{sort}(P)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \text{cp } P(f) \text{sort}(P) \text{filterp}$
 combining the arguments' sorted lists by listcp , mapping f and filtering is below sorting the f -images and filtering



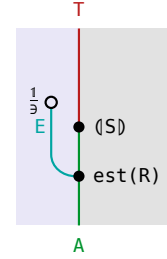
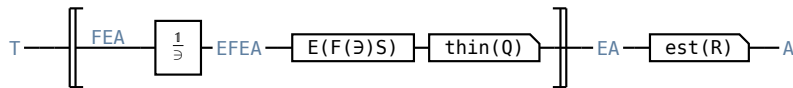
- (a) $\text{sort}(fP f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(P)$ — (8.3b)
 (b) $FP \sqsubseteq fP f^\circ$ — mon-str, $\triangleq$ thinlist
 (c) $F(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(FP)$ — (8.3b)

$\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \subseteq R \wedge 1 \in Q \wedge Q \subseteq Q \wedge R \circ R \subseteq R \wedge \text{MonoAlg}(f_1 \text{ } p_1, Q) \wedge \text{MonoAlg}(f_2 \text{ } p_2, Q)$ Theorem 8.2
 $\wedge (\forall X. \forall Y. X \subseteq Y \Rightarrow \text{sortF}(X) \subseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \text{sortF}(f_1 \text{ } p_1) \subseteq \text{listf}_1 \subseteq P(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 \text{ } p_2) \subseteq \text{listf}_2 \subseteq P(f_2) \text{sort}(P) \wedge$
 $\text{sort}(P) \text{filter} p_1 \subseteq E(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filter} p_2 \subseteq E(p_2) \text{sort}(P) \wedge F(\text{sort}(P)) \text{listcp} \subseteq \text{cp}$
 $\text{sortF}(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \subseteq \text{cup } \text{sort}(P) \wedge$
 $\text{sort}(P) \text{thinlist}(Q) \subseteq \text{thin}(Q) \text{sort}(P) \wedge \text{sort}(P) \text{minlist}(R) \subseteq \text{est}(R) \wedge g_1 = \text{listf}_1 \text{ filter } p_1 \wedge$
 $g_2 = \text{listf}_2 \text{ filter } p_2 \wedge S = f_1 \text{ } p_1 \cup f_2 \text{ } p_2 \Rightarrow (\text{listcp}(g_1, g_2) \text{ merge}(P) \text{ thinlist}(Q))$
 $\text{minlist}(R) \subseteq \frac{(S)}{S} \text{ est}(R)$
 a fold on sorted lists of partial solutions, thinned at every step — at $\triangleq \text{thinlist}$'s binary thinning data.

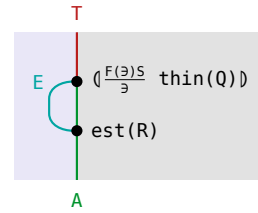
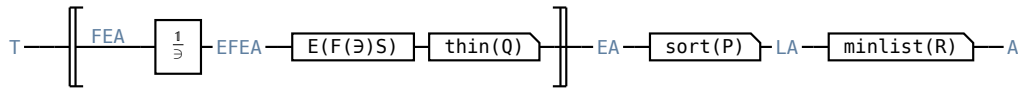
circuit



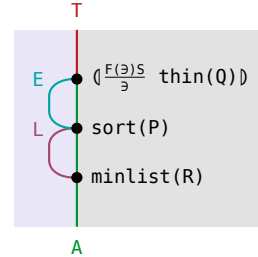
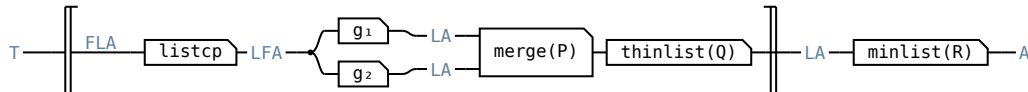
Hinze–Marsden


 $\sqsubseteq$


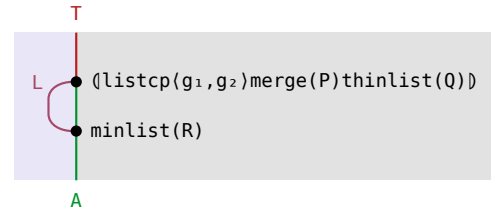
Corollary 8.1 at $f_1 p_1$ and $f_2 p_2$ monotonic on $Q \triangleq \text{thinlist}$


 $\sqsubseteq$


$\text{sort}(P) \text{ minlist}(R) \subseteq \text{est}(R)$ — (8.3b) at its $Q \triangleq R$


 $\sqsubseteq$


fusion at (8.3c)

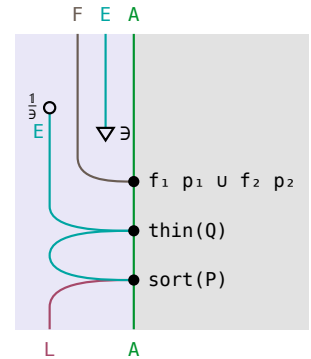
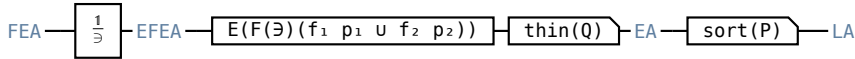


$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, P) \wedge \text{MonoAlg}(f_2, P) \wedge \quad (8.3c) \\
& \text{sortF}(f_1 \text{ Pf}_1^\circ) \text{listf}_1 \sqsubseteq \text{P}(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 \text{ Pf}_2^\circ) \text{listf}_2 \sqsubseteq \text{P}(f_2) \text{sort}(P) \wedge \\
& \text{sort}(P) \text{filterp}_1 \sqsubseteq \text{E}(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filterp}_2 \sqsubseteq \text{E}(p_2) \text{sort}(P) \wedge \text{F}(\text{sort}(P)) \text{listcp} \sqsubseteq \text{cp} \text{sortF}(\text{F}(P)) \\
& \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \wedge \text{sort}(P) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(P) \Rightarrow \\
& \text{F}(\text{sort}(P)) \text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \text{merge}(P) \text{thinlist}(Q) \sqsubseteq \frac{\text{F}(\exists)(f_1 \text{ p}_1 \cup f_2 \text{ p}_2)}{\exists} \\
& \text{thin}(Q) \text{sort}(P)
\end{aligned}$$

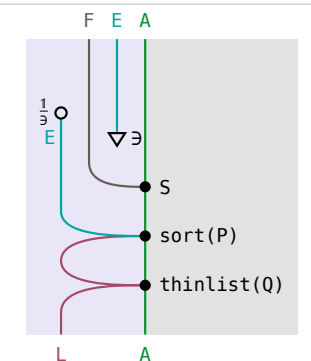
sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit

Hinze–Marsden



$$\begin{aligned}
& \text{FEA} \text{---} \boxed{\frac{1}{3}} \text{---} \text{EFEA} \text{---} \boxed{\text{E}(\text{F}(\exists)S)} \text{---} \text{EA} \text{---} \boxed{\text{sort}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& \text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P) \text{ --- (8.3b)}
\end{aligned}$$



$$\begin{aligned}
& \text{FEA} \text{---} \boxed{\frac{1}{3}} \text{---} \text{EFEA} \text{---} \boxed{\text{E}(\text{F}(\exists)f_1 \text{ p}_1)} \text{---} \text{EA} \text{---} \boxed{\text{cup}} \text{---} \text{EA} \text{---} \boxed{\text{sort}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& = \\
& \left\langle \frac{\text{F}(\exists)f_1 \text{ p}_1}{\exists}, \frac{\text{F}(\exists)f_2 \text{ p}_2}{\exists} \right\rangle \text{cup} \text{sort}(P) \text{ thinlist}(Q) \\
& S = (f_1 \text{ p}_1) \cup (f_2 \text{ p}_2) \text{ --- } \triangleq \text{thinlist, then } \triangleq \text{cup}
\end{aligned}$$

$$\begin{aligned}
& \text{FEA} \text{---} \boxed{\frac{1}{3}} \text{---} \text{EFEA} \text{---} \boxed{\text{E}(\text{F}(\exists)f_1 \text{ p}_1)} \text{---} \text{EA} \text{---} \boxed{\text{sort}(P)} \text{---} \boxed{\text{merge}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& \text{FEA} \text{---} \boxed{\frac{1}{3}} \text{---} \text{EFEA} \text{---} \boxed{\text{E}(\text{F}(\exists)f_2 \text{ p}_2)} \text{---} \text{EA} \text{---} \boxed{\text{sort}(P)} \text{---} \boxed{\text{merge}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& \left\langle \frac{\text{F}(\exists)f_1 \text{ p}_1}{\exists}, \frac{\text{F}(\exists)f_2 \text{ p}_2}{\exists} \right\rangle (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \text{ thinlist}(Q) \\
& (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \sqsubseteq \text{cup} \text{sort}(P) \text{ --- (8.3b)}
\end{aligned}$$

$$\begin{aligned}
& \text{FEA} \text{---} \boxed{\frac{1}{3}} \text{---} \text{EFEA} \text{---} \boxed{\text{E}(\text{F}(\exists)f_1 \text{ p}_1)} \text{---} \text{EA} \text{---} \boxed{\text{sort}(P)} \text{---} \text{LA} \text{---} \boxed{\text{merge}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& \text{FEA} \text{---} \boxed{\frac{1}{3}} \text{---} \text{EFEA} \text{---} \boxed{\text{E}(\text{F}(\exists)f_2 \text{ p}_2)} \text{---} \text{EA} \text{---} \boxed{\text{sort}(P)} \text{---} \text{LA} \text{---} \boxed{\text{merge}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& \left\langle \frac{\text{F}(\exists)f_1 \text{ p}_1}{\exists} \text{sort}(P), \frac{\text{F}(\exists)f_2 \text{ p}_2}{\exists} \text{sort}(P) \right\rangle \text{merge}(P) \text{ thinlist}(Q) \\
& \langle X, Y \rangle (\text{sort}(P) \times \text{sort}(P)) = \langle X \text{ sort}(P), Y \text{ sort}(P) \rangle \text{ --- bdm-prod-laws}
\end{aligned}$$

$$\begin{aligned}
& \text{FEA} \text{---} \boxed{\text{F}(\text{sort}(P))} \text{---} \text{FLA} \text{---} \boxed{\text{listcp}} \text{---} \text{LFA} \text{---} \boxed{\text{listf}_1} \text{---} \boxed{\text{filterp}_1} \text{---} \text{LA} \text{---} \boxed{\text{merge}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& \text{FEA} \text{---} \boxed{\text{F}(\text{sort}(P))} \text{---} \text{FLA} \text{---} \boxed{\text{listcp}} \text{---} \text{LFA} \text{---} \boxed{\text{listf}_2} \text{---} \boxed{\text{filterp}_2} \text{---} \text{LA} \text{---} \boxed{\text{merge}(P)} \text{---} \boxed{\text{thinlist}(Q)} \text{---} \text{LA} \\
& \text{F}(\text{sort}(P)) \text{ listcp} \langle g_1, g_2 \rangle \text{merge}(P) \text{ thinlist}(Q) \\
& \text{Lemma 8.1 at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then } X \langle g_1, g_2 \rangle \sqsubseteq \langle Xg_1, Xg_2 \rangle \text{ --- bdm-prod-laws; } g_i \triangleq \text{list}(f_i) \\
& \text{filter}(p_i) \text{ --- } \triangleq \text{thinlist}
\end{aligned}$$

§8.4. The knapsack problem

(8.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}, \text{ value} \triangleq \text{list}(\text{vol}) \text{ sum}, \text{ weight} \triangleq \text{list}(\text{wt}) \text{ sum}.$

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2]) \circ \text{w}$, within w the coreflexive on xs with weight $\text{xs} \leq w$, $0 \leq w$.

$R \triangleq \text{value} \geq \text{value}^\circ$, $Q \triangleq \text{Rn}(\text{weight} \leq \text{weight}^\circ)$, $P \triangleq R \circ \text{w}$.

$\text{FA} = 1 + \text{Item} \times \text{A}$, $\text{listcp} = \text{wrap} + \text{cpr}$, $g_1 \triangleq \text{list}([\text{nil}, \text{cons}])$ $\text{filter}(\text{within } w) = [\text{list}(\text{nil}), h_1]$,
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2].$

$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w)$, $h_2 \triangleq \text{list}(\pi_2).$

§8.5. The paragraph problem

(8.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[a]#xs, glue(a,xs)=[a]#head(xs)#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(8.5b)

the law	what it says
$(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
$(1 \times Q) \text{ new} \sqsubseteq \text{new } Q$ $(1 \times Q) (\text{glue } (ok \ w)) \sqsubseteq \text{glue } (ok \ w) Q$ $\text{cons monotonic on } \text{collect} \sqsubseteq \text{collect}^\circ, .$	both halves are monotonic on Q once ties in waste are broken by the first line
merge $T = \text{cat}$; $P \triangleq \text{head prefix head}^\circ$ also serves	T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input

(8.5c)

$(\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap}$
 $\text{wrap,new}]T[\text{wrap wrap,new}]^\circ) \text{listf}_1 \sqsubseteq P([\text{wrap wrap,new}]) \text{sortP} \wedge \text{sortF}([\text{wrap wrap,glue}]T[\text{wrap}$
 $\text{wrap,glue}]^\circ) \text{listf}_2 \sqsubseteq P([\text{wrap wrap,glue}]) \text{sortP} \wedge \text{sortP filterp}_2 \sqsubseteq E(ok) \text{sortP} \wedge F(\text{sortP}) \text{listcp} \sqsubseteq \text{cpr}$
 $\text{sortF}(F(T)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup sortP} \wedge \text{sortP thinlist} \sqsubseteq \text{thin}(Q(\text{len},w)) \text{sortP} \wedge \text{sortP}$
 $\text{minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp}(\text{listf}_1, \text{listf}_2 \text{ filterp}_2) \text{ mergeP thinlist}] \text{minlist} \sqsubseteq \frac{\text{partition fits}}{3} \text{est}(R))$
 a paragraph laid out as a fold that thins the layouts kept at each word

circuit	Hinze–Marsden
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^*(\text{Word})) \xrightarrow{E(\text{partition fits})} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $\frac{\text{partition list}^*(\text{fits } w)}{3} \text{est}(R)$	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap,new} u [\text{wrap wrap,glue} ok])} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $= \frac{([\text{wrap wrap,new} u (\text{glue } (ok \ w))])}{3} \text{est}(R)$ fusion, every word fits on a line by itself.	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{3}} E(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap,new} u [\text{wrap wrap,glue} ok])} E(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $= \frac{([\text{wrap wrap,new} u ([\text{wrap wrap,glue} (ok \ w)])])}{3} \text{est}(R)$ the algebra as $(f_1 p_1) \cup (f_2 p_2)$, $p_1 \triangleq 1 \text{ — } \triangleq \text{thinlist}$.	
$\text{listcp } (g_1, g_2) \text{ cat thinlist}(Q) \text{ minlist}(R)$ binary thinning theorem, at $P \triangleq T$ with merge $T = \text{cat}$, Q from (8.5b)	
$([\text{start, cpr } (h_1, h_2) \text{ cat thinlist}(Q)])$ $= \text{minlist}(R)$ $\text{listcp} = \text{wrap} + \text{cpr}$, g_i along the coproduct — $\triangleq \text{partition}$	

§8.6. Bitonic tours

(8.6a)

$FA = (City \times City) + (City \times A)$, the base functor of cons-lists of length at least two; $listcp = wrap + cpr$; $tc : City \times City \rightarrow Real$, neither positive nor symmetric.

$start(a, b) = ([a, b], [a, b])$, $dropl(a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys)$, $dropr(a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys)$, $tour \triangleq ([start, dropl \cup dropr])$.

$cost(xs, ys) = outcost\ xs + incost\ ys$, $outcost[a_0, \dots, a_n] = tc(a_0, a_1) + \dots + tc(a_{n-1}, a_n)$, $incost[a_0, \dots, a_n] = tc(a_1, a_0) + \dots + tc(a_n, a_{n-1})$.

$next \triangleq tail\ head$, $next2 \triangleq next \times next$, $head2 \triangleq head \times head$, $R \triangleq cost \leq cost^\circ$, $Q \triangleq Rn(next2\ next2^\circ) \cap (head2\ head2^\circ)$, $P \triangleq T$, $g_1 \triangleq list([start, dropl])$, $g_2 \triangleq list([start, dropr])$.

(8.6b)

the law	what it says
$(1 \times R)\ dropl \sqsubseteq dropl\ R\ \text{FALSE}$	neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists
$(1 \times R)\ dropr \sqsubseteq dropr\ R\ \text{FALSE}$	
$(1 \times Q)\ dropl \sqsubseteq dropl\ Q$	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input
$(1 \times Q)\ dropr \sqsubseteq dropr\ Q$	

(8.6c)

$(\forall X. \forall Y. X \sqsubseteq Y \Rightarrow sortF(X) \sqsubseteq sortF(Y)) \wedge$ $sortF([start, droplFn] \tau [start, droplFn]^\circ) g_1 \sqsubseteq P([start, droplFn]) sortP \wedge$ $sortF([start, droprFn] \tau [start, droprFn]^\circ) g_2 \sqsubseteq P([start, droprFn]) sortP \wedge F(sortP) listcp \sqsubseteq cp$ $sortF(F(\tau)) \wedge (sortP \times sortP) cat \sqsubseteq cup\ sortP \wedge sortP\ thinlist \sqsubseteq thin(Qc(tc)) sortP \wedge sortP$ $minlist \sqsubseteq est(R) \Rightarrow (listcp(g_1, g_2) cat\ thinlist) minlist \sqsubseteq \frac{(tourAlg)}{\exists} est(R)$ a least-cost bitonic tour, as a fold that thins the tours kept at each city	
circuit	Hinze–Marsden
$dCL(City \times City, City) \xrightarrow{\frac{1}{\exists}} E(dCL(City \times City, City)) \xrightarrow{E(tourAlg)} E(Journey^2) \xrightarrow{est(R)} Journey$ $\frac{tour}{\exists} est(R)$	
$dCL(City \times City, City) \xrightarrow{\frac{1}{\exists}} E(dCL(City \times City, City)) \xrightarrow{E([start, droplFn] \cup [start, droprFn])} E(Journey^2) \xrightarrow{est(R)} Journey$ $= \frac{([start, dropl \cup dropr])}{\exists} est(R)$ $tour \triangleq ([start, dropl \cup dropr]) \triangleq tour$	
$dCL(City \times City, City) \xrightarrow{\frac{1}{\exists}} E(dCL(City \times City, City)) \xrightarrow{E([start, droplFn] \cup [start, droprFn])} E(Journey^2) \xrightarrow{est(R)} Journey$ $\sqsubseteq (listcp(g_1, g_2) cat\ thinlist(Q)) minlist(R)$ binary thinning theorem, at $P \triangleq T$ with merge $\tau = cat$, Q from (8.6b)	
$([start\ wrap, cpr\ (list(dropl), list(dropr))\ cat\ thinlist(Q)]) minlist(R)$ $listcp = wrap + cpr$, $g_i = [list(start), list(drop_i)] \triangleq tour$; quadratic, two tours added per step	